

MVP OVERVIEW / AUGUST 2026

CA

ChargeAssert

Financial release testing
for EV charging.

Catch billing defects before they reach customers or
revenue systems.

SELIM ABOULEILA

Independent public portfolio project
Public and synthetic data only

01 / MVP OVERVIEW

The product

ChargeAssert is a test harness for EV-charging software. Before a new version goes live, it replays realistic charging sessions and checks that every session produces the correct financial result.



A release can look technically healthy while this financial chain is wrong.

What it catches

- + Missing charging records
- + Duplicate customer charges
- + Incorrect energy or duration
- + Wrong tariff or tariff version
- + Retry and idempotency errors
- + Late or out-of-order events

PROTECT CUSTOMERS

Stop duplicate and inflated charges before deployment.

PROTECT REVENUE

Expose missing or understated billable records.

MOVE FASTER

Give engineering a reproducible cause, not just an error count.

THE BUSINESS VALUE

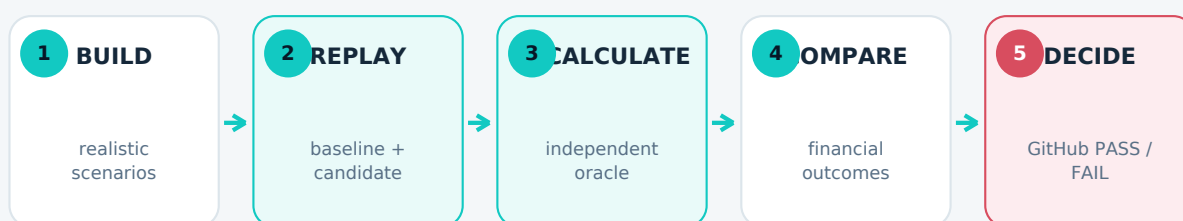
Shift financial defect discovery from post-production reconciliation to a clear release decision before deployment.

Primary users: QA and platform engineering, revenue assurance, OCPP / OCPI integration teams, and charging-platform vendors.

02 / MVP OVERVIEW

How it works

ChargeAssert sends the same deterministic scenario manifest through a trusted baseline and the candidate release, then compares both outputs with an independent expected result.

ONE MANIFEST. TWO RELEASES. ONE FINANCIAL VERDICT.

The same seed, event clock and inputs make every failure reproducible.

BASELINE VS CANDIDATE

Both releases receive identical events, IDs and timestamps. The comparison exposes what changed in the candidate.

INDEPENDENT ORACLE

Expected energy, tariff and cost are calculated separately. An old baseline defect cannot silently become acceptable.

Black-box by design

The MVP tests controlled mock systems. A future customer can replace the mock adapter with approved staging endpoints. ChargeAssert needs inputs, outputs and tariff rules - not access to proprietary source code.

WHY NO MACHINE LEARNING?

The MVP uses deterministic rules because financial checks need known expected outcomes, clear explanations and exact reproduction. ML can be explored later with labelled operational data.

03 / MVP OVERVIEW

A focused MVP

The first release proves one complete financial quality gate. Everything else stays outside the boundary until the core workflow is reliable and useful.

INCLUDED

- + Synthetic OCPP and OCPI-shaped events
- + Baseline and candidate comparison
- + Six flagship failure scenarios
- + Independent financial assertions
- + Modeled EUR exposure
- + GitHub PASS / FAIL report
- + Reproducible Databricks pipeline

NOT IN THE MVP

- + Real card or payment processing
- + Bank, PSP, ERP or settlement integration
- + Full OCPP / OCPI certification
- + Production monitoring or recovery
- + Every tariff, tax system or currency
- + Machine learning
- + Confidential operator data

MVP outputs

VERDICT

GitHub PASS /
FAIL

CAUSE

First divergence

IMPACT

Overbilling vs
leakage

EVIDENCE

Reproducible run
manifest

DEFINITION OF DONE

A deliberately faulty candidate fails, the report explains why, and the corrected candidate passes the exact same scenario manifest.

04 / MVP OVERVIEW

Six failures that matter

The scenario library stays intentionally small. Each case represents a defect with a clear customer or revenue consequence and a deterministic expected outcome.

01 MISSING CDR

A completed session never becomes billable.

02 DUPLICATE CHARGE

A retry creates a second financial record.

03 WRONG ENERGY

Meter order, reset or units distort kWh.

04 WRONG TARIFF

The incorrect effective price is applied.

05 RETRY FAILURE

An HTTP error breaks idempotency.

06 LATE EVENT

Out-of-order data changes the final state.

Core financial checks

- + One completed billable session produces one final CDR.
- + CDR energy matches validated meter evidence.
- + The correct effective tariff is applied.
- + The final amount matches the independent calculation.

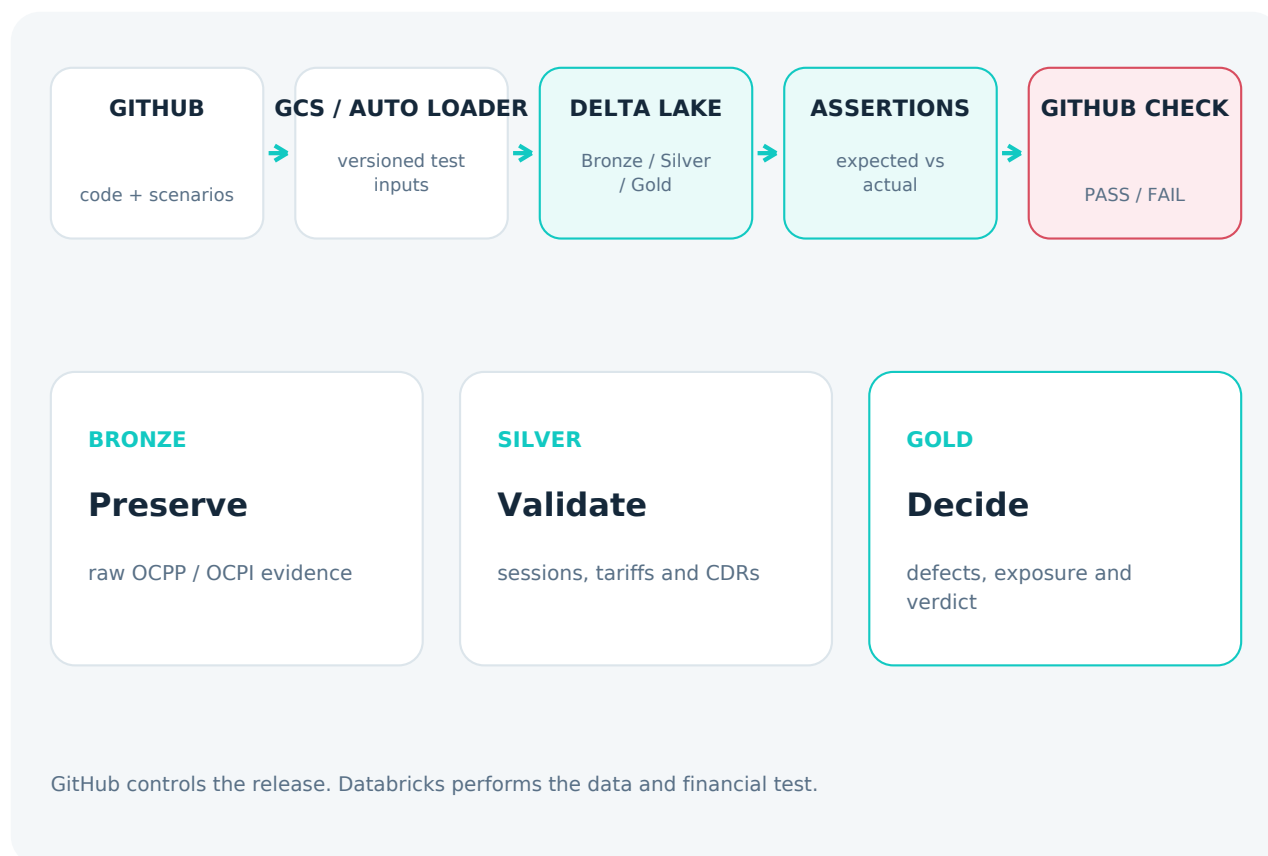
**ONE
IMPORTANT
DISTINCTION**

The same event ID delivered twice is a transport duplicate to deduplicate. Two different CDR IDs for one session are a business defect to preserve and report.

05 / MVP OVERVIEW

A simple Databricks architecture

GitHub is the control plane. Databricks is the execution plane. Public-data snapshots and synthetic events move through a compact lakehouse flow and return one release verdict.



INTERFACES

- + OCPP subset for charger-to-platform events
- + OCPI-aligned Sessions, Tariffs and CDRs
- + GitHub Actions and Declarative Automation Bundles

PUBLIC INPUTS

- + ACN-Data for realistic charging behavior
- + French IRVE data for static station profiles
- + Synthetic tariffs and controlled fault injection

DATA PROVENANCE

ACN behavior and French IRVE station metadata are combined synthetically. They are not presented as real French charging transactions or actual operator tariffs.

06 / MVP OVERVIEW

The release gate

The product moment is a pull request that fails for a financial reason, shows the first divergence and turns green after the exact defect is fixed.

chargeassert / release-gate

RELEASE GATE: FAILED

Sessions tested	50,000
Missing CDR delta	+0.42%
Duplicate charges	+0.18%
Modeled monthly exposure	EUR 24,380

FIRST DIVERGENCE

An OCPI retry after HTTP 500 created a second CDR for the same logical charging session.

Baseline: 8f6c2a1
Candidate: b319ad4
Scenario: retry_after_500

Illustrative result. Counts and financial figures are modeled, not actual losses.

What the report must answer

WHAT BROKE?

The first event, retry or tariff decision where behavior diverged.

WHO IS EXPOSED?

Customer overbilling and operator underbilling shown separately.

CAN WE REPLAY IT?

Run ID, seed, SHAs, snapshot hashes and a reproduction command.

MODELED EXPOSURE

defect-rate delta x assumed monthly sessions x assumed impact per affected session

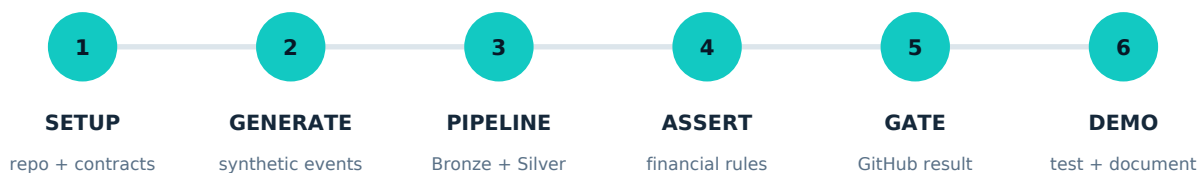
All projections show their assumptions. They are modeled exposure - never actual loss, recovered revenue or proven savings.

07 / MVP OVERVIEW

Build the MVP in six weeks

The first six weeks deliver a complete portfolio demonstration. The next phase should be driven by a real staging partner, not by adding complexity for its own sake.

SIX-WEEK MVP



NEXT PHASE

Connect one consenting staging system, import real test tariff contracts, extend the evidence pack and measure false positives and investigation time.

HONEST BOUNDARIES

- + Not an OCPP / OCPI certification authority
- + Not a card-payment or settlement platform
- + Not proof of real financial savings

SOURCE SET

Open Charge Alliance - OCPP
Official OCPI repository
Caltech ACN-Data
French national IRVE dataset
Databricks Auto Loader
Databricks medallion architecture

**THE MVP
PROMISE**

Catch missing, duplicate and incorrectly priced EV-charging records before the associated release reaches production.

Independent educational and portfolio project. No confidential employer, operator or customer data. All monetary outcomes are synthetic and modeled.